Object Oriented Programming (CSEG1044)

Unit 01 – Lecture 02: Java Overview + Program Structure + Features

April 1, 2026

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Lecture Snapshot

- **Unit focus:** Introduction to OOPs and Java Fundamentals

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Lecture Snapshot

- **Unit focus:** Introduction to OOPs and Java Fundamentals
- **Course thread:** Use Java Overview + Program Structure + Features to make the first text-based version of Campus Resource Manager easier to reason about before the full object model arrives.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Recall Before You Start

- **OOP vocabulary**: The previous lecture introduced objects, classes, and modelling; this lecture connects them to the Java platform. Main reference: `Unit 01 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Recall Before You Start

- **OOP vocabulary**: The previous lecture introduced objects, classes, and modelling; this lecture connects them to the Java platform. Main reference: `Unit 01 / Lecture 01 slides/notes`.
- **Problem to program flow**: Once the model is clear, the next question is how Java compiles and runs it. Main reference: `Unit 01 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Learning Outcomes

- Explain JDK vs JRE vs JVM and what bytecode is.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Learning Outcomes

- Explain JDK vs JRE vs JVM and what bytecode is.
- Compile and run a Java program; recognize compile-time vs runtime errors.

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Learning Outcomes

- Explain JDK vs JRE vs JVM and what bytecode is.
- Compile and run a Java program; recognize compile-time vs runtime errors.
- List key features of Java and their practical meaning.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Key Terms to Know

- **JDK (Java Development Kit)**: tools for development (compiler `javac`, docs, etc.).
- **JRE (Java Runtime Environment)**: runtime needed to run Java apps (JVM + libraries).
- **JVM (Java Virtual Machine)**: executes Java bytecode (`.class`).
- **Bytecode**: platform-neutral instructions produced by `javac`.
- **Classpath**: where the JVM looks for compiled classes and libraries.
- **Compile-time error**: detected by the compiler before running.
- **Runtime error/exception**: occurs while the program is executing.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics);

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Java ecosystem: JDK vs JRE vs JVM* → *Bytecode and the compile/run flow* → *Basic program structure: class + `main`* → *Compile-time vs runtime errors*.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

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3. Trace the worked example and connect each line back to one concept frame.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

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3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Unit 01: Introduction to OOPs

- Lecture focus: Java Overview + Program Structure + Features

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Today: Roadmap

- Java ecosystem: JDK/JRE/JVM, bytecode, compile/run flow

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Today: Roadmap

- Java ecosystem: JDK/JRE/JVM, bytecode, compile/run flow
- Basic program structure: class, ‘main‘, compilation errors vs runtime errors

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Today: Roadmap

- Java ecosystem: JDK/JRE/JVM, bytecode, compile/run flow
- Basic program structure: class, ‘main‘, compilation errors vs runtime errors
- Features of Java (platform independence, security, GC overview)

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

JDK vs JRE vs JVM (picture)

JDK
developer tools + JRE (includes `javac`)

JRE
JVM + standard libraries (run-time)

JVM
loads + verifies + executes bytecode (`.class`)

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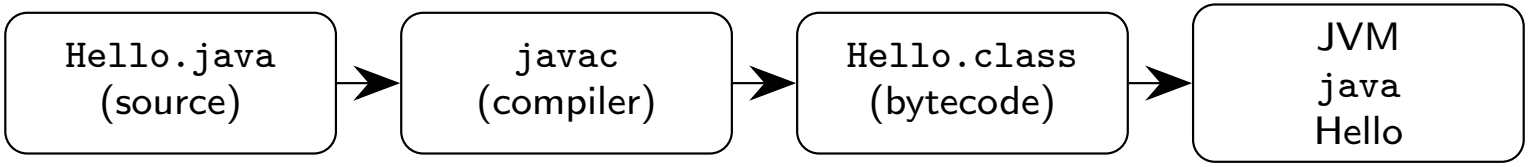
Compile → Run flow



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Program structure (minimum)

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello, Java!");
    }
}
```

- If the class is `public`, file name must be `Hello.java`.

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Program structure (minimum)

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public class Hello {
    public static void main(String[] args) {
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```

- If the class is `public`, file name must be `Hello.java`.
- JVM starts execution from `main`.

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Compile-time vs Runtime (quick)

- **Compile-time error:** caught by `javac` (syntax/type errors).

Examples

- Compile-time: missing semicolon, "cannot find symbol"
- Runtime: `ArithmeticException`, `NullPointerException`

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Compile-time vs Runtime (quick)

- **Compile-time error:** caught by `javac` (syntax/type errors).
- **Runtime exception:** program compiles but fails while running.

Examples

- Compile-time: missing semicolon, "cannot find symbol"
- Runtime: `ArithmeticException`, `NullPointerException`

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Features of Java (what you get)

- Platform independent: bytecode + JVM

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Features of Java (what you get)

- Platform independent: bytecode + JVM
- Strongly typed: many errors caught early

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Features of Java (what you get)

- Platform independent: bytecode + JVM
- Strongly typed: many errors caught early
- Garbage collection: automatic memory management

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Features of Java (what you get)

- Platform independent: bytecode + JVM
- Strongly typed: many errors caught early
- Garbage collection: automatic memory management
- Rich libraries: collections, I/O, GUI, DB, networking

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Features of Java (what you get)

- Platform independent: bytecode + JVM
- Strongly typed: many errors caught early
- Garbage collection: automatic memory management
- Rich libraries: collections, I/O, GUI, DB, networking
- Security model: class loading + bytecode verification

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Checkpoint and Reflection

Reflection prompt

Where would *Java Overview + Program Structure + Features* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Java ecosystem: JDK vs JRE vs JVM*, *Bytecode and the compile/run flow*, the worked example, or the recap?

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Demo: javac and java

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello, Java!");
    }
}
```

- Compile: `javac Hello.java`

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

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public class Hello {
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- Compile: `javac Hello.java`
- Run: `java Hello`

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Demo: javac and java

```
public class Hello {
    public static void main(String[] args) {
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}
```

- Compile: `javac Hello.java`
- Run: `java Hello`
- Try: change message and re-run.

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Demo: runtime exception (observe)

```java
public class DivideDemo {
    public static void main(String[] args) {
        int a = 10, b = 0;
        System.out.println(a / b);
    }
}
```

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Mini Exercises (2–3 mins)

1. Which contains `javac`: JDK, JRE, or JVM?

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Mini Exercises (2–3 mins)

1. Which contains `javac`: JDK, JRE, or JVM?
2. If you write `public class Student`, what should the file be named?

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Mini Exercises (2–3 mins)

1. Which contains `javac`: JDK, JRE, or JVM?
2. If you write `public class Student`, what should the file be named?
3. Is "cannot find symbol" compile-time or runtime?

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Watch-outs

- Using JDK, JRE, and JVM as interchangeable terms.

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Watch-outs

- Using JDK, JRE, and JVM as interchangeable terms.
- Forgetting that a public class name must match the file name.

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Watch-outs

- Using JDK, JRE, and JVM as interchangeable terms.
- Forgetting that a public class name must match the file name.
- Mixing compile-time errors with runtime failures.

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Key Takeaways

- JDK is for development (includes `javac`); JRE/JVM are for running.

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- JDK is for development (includes `javac`); JRE/JVM are for running.
- Java compiles to bytecode (`.class`) and runs on the JVM.

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Key Takeaways

- JDK is for development (includes `javac`); JRE/JVM are for running.
- Java compiles to bytecode (`.class`) and runs on the JVM.
- Compile-time errors stop compilation; runtime exceptions happen while running.

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Checkpoint Questions

1. Why does Java use bytecode?

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Checkpoint Questions

1. Why does Java use bytecode?
2. Give one compile-time error example and one runtime exception example.

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Checkpoint Questions

1. Why does Java use bytecode?
2. Give one compile-time error example and one runtime exception example.
3. What is the role of `main`?

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.

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- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

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Next Lecture Preview

Next: I/O + Command Line + Types + Operators

- Practice running a program with command line arguments.

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Next Lecture Preview

Next: I/O + Command Line + Types + Operators

- Practice running a program with command line arguments.
- Revise basic numeric types and operator precedence.

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